Venkata Lakshmi Parimala Pasupuleti

AI Engineer · AI Evaluation & Model Risk · Python / LangGraph / LLMOps · Enterprise AI Safety at Scale

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SUMMARY

AI Engineer with 9 years of production software ownership. Built AgentProbe -- an open-source AI evaluation and model risk platform: CI-integrated regression gate (quality drift detected before deployment, not discovered in production), LLM-as-judge scoring with model-level groundedness and hallucination tracking, token efficiency monitoring, and multi-provider LLM routing for model swap on deprecation. Core data modeling and data analysis background underpins every evaluation pipeline. Built to do exactly what T-Mobile's AI Evaluation & Model Risk role describes: give teams evidence to trust -- or not trust -- each model and agent before it runs in production. Built LeaseI -- an 8-agent LangGraph system with RAG, prompt-engineered guardrails, and HITL on regulated decisions. Regulated-domain production: AI on FHIR (HIPAA-compliant, PHI, RBAC) + Entrata PHI compliance (zero incidents). Seven years shipping enterprise Python systems at 100K+ MAU scale. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI Evaluation & Model Risk: LLM Evaluation, Evaluation Harness, Model Risk Assessment, Evaluation Pipelines, CI/CD Eval Gate, LLM-as-Judge, Pass@k Harness, Agent Tracing, Token Efficiency Monitoring, Quality Drift Detection, Hallucination Detection, Groundedness Scoring, Model Cards, Regression Suites, Production Monitoring, LLMOps, MLOps

AI & LLM Engineering: Python, LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, RAG Pipelines (Retrieval-Augmented Generation), Vector Databases, ChromaDB, Embeddings, Semantic Search, Prompt Engineering, Fine-tuning (HuggingFace Transformers, PyTorch), Tool Calling, Guardrails, Output Validation, HITL, Structured Outputs, OpenAI API, Anthropic API, Groq, Ollama, Multi-provider LLM routing, MCP

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes (K8s), CI/CD (GitHub Actions), FastAPI, REST APIs, RabbitMQ, Redis, Microservices, HIPAA Compliance, PHI Data Handling, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum

Languages: Python, JavaScript, TypeScript, Java, SQL, PL/SQL, React.js, Angular, Flask, Bash, HTML/CSS

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built a production LangGraph agentic system with RAG, prompt-engineered guardrails, and output validation; ran model selection and orchestration architecture decisions for external client delivery.

Entrata (PropTech SaaS · 100K+ MAU · PHI compliance · enterprise production)

Sep 2018 - Aug 2024

Senior Software Engineer

- **Owned compliance-critical data systems in a regulated enterprise environment** -- annual HUD income-limit pipeline (180,000 domain records, housing eligibility for 100K+ residents); zero compliance violations across 7 years; baseline validation, reconciliation, and rollback preparation built into every release cycle.
- Led AWS S3 document migration -- 100% PHI compliance, Libsodium-encrypted pipelines, zero security incidents; owned HIPAA-adjacent data workflows throughout tenure.
- Shipped 20+ full-stack production modules (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; built Python/Bash automation pipelines eliminating 80% of manual workload; optimized 200+ SQL queries (30% performance gain).
- Mentored 5+ junior engineers through cross-functional collaboration; conducted 300+ peer code reviews; built internal debugging tools reducing investigation time by 40%.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Fine-tuned a custom classification model architecture using HuggingFace Transformers and PyTorch; deployed inference API on Quart; presented model design, training methodology, and evaluation results to faculty.
- Built multi-source prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest anomaly detection across 3 research pipelines; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: AI Evaluation Harness & Model Risk Platform

2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing

- Open-source AI evaluation and model risk platform: CI-integrated regression gate detecting quality drift before deployment, LLM-as-judge scoring with model-level groundedness and hallucination tracking (model cards in practice), token efficiency monitoring flagging burn-rate anomalies, multi-provider LLM routing enabling safe model swap on deprecation. Built to give teams evidence -- not guesswork -- on which models and agents are safe to keep running.

- 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); 15-endpoint FastAPI backend; multi-provider routing (Anthropic, OpenAI, Groq, Ollama); evaluated LeaseI across 1,240+ production-quality test cases.

Leasel: 8-Agent LangGraph System with RAG & HITL Guardrails

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit

- Production agentic system: 8-agent LangGraph with RAG over compliance documents, prompt-engineered output validation, HITL escalation on regulated decisions, conditional routing. 93.6% precision across 1,240 test cases -- operated safely via AgentProbe's evaluation harness.

AI on FHIR: HIPAA-Compliant Medical Records Query System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR · spaCy · BERT · Docker

- HIPAA-compliant AI system over structured PHI data -- RBAC access controls, responsible AI data governance, LLM + BERT integration; reduced manual clinical retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017